



Plant spectral diversity integrates functional and phylogenetic components of biodiversity and predicts ecosystem function.

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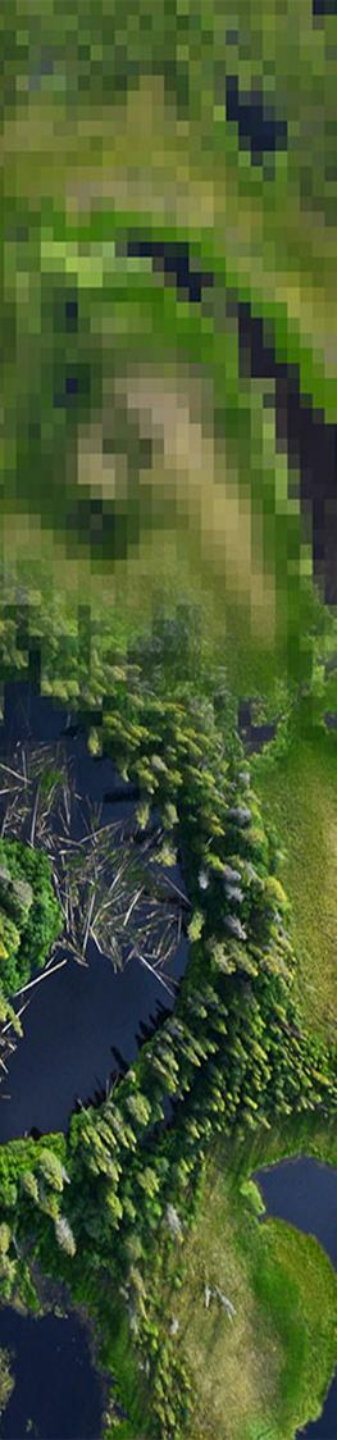
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Background

Proposed concept; to be operationalized in the paper

Plant *spectral diversity* integrates **functional** and **phylogenetic** components of **biodiversity** and predicts **ecosystem function**.

Biodiversity → the variety of living things

the processes which maintain terrestrial life

Phylogenetic Diversity → *evolutionary divergence*

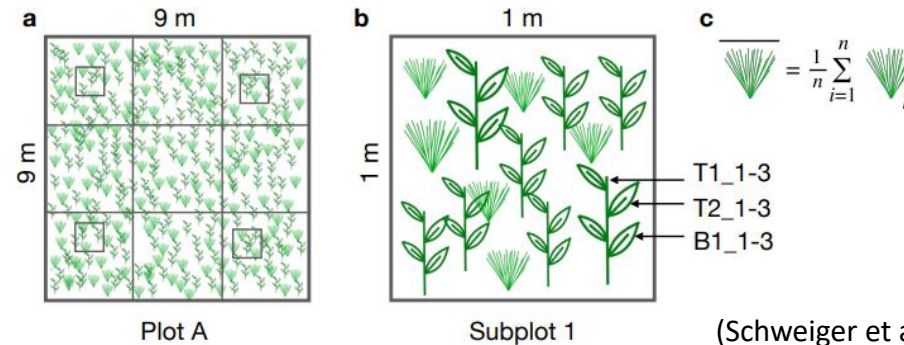
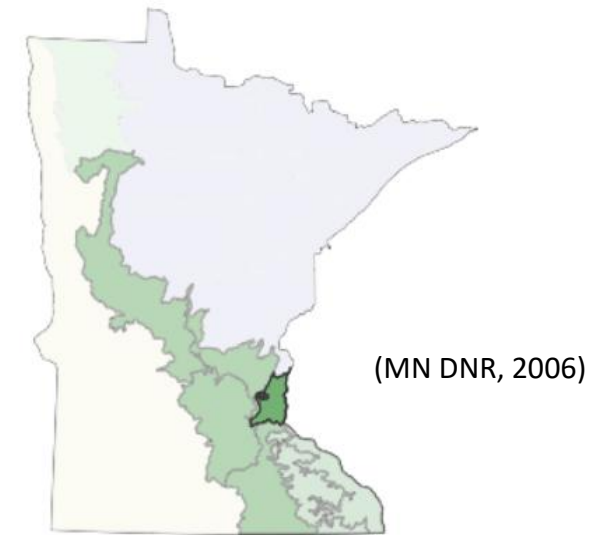
(Trivedi et al., 2018)

Functional Diversity → *difference in roles / niches*

Data Collection



Cedar Creek Ecosystem Science Reserve, East Bethel, MN, USA
Eco Subsection: St. Paul-Baldwin Plains and Moraines (Tallgrass Prairie)



Measurement	Processing Method	Value
Spectrometry	Calibrate, resample, and plot	Spectral Profiles
Hyperspectral Imagery		
Leaf Samples	Chemical Analysis	Functional Traits
	Follow Davies et al. (2016)	Phylogenetic Traits
Above Ground Biomass	Equate with Net Primary Productivity	Ecosystem Function: Productivity

Methods: Dissimilarity and Diversity

$${}^qD(TM) = 1 + (S-1) \times {}^qE(T) \times M'$$

Diversity:
Dispersion from
the Trait Mean

M' = mean dispersion
= average distance among
all possible species pairs

${}^qE(T)$ = evenness of traits
= the extent to which
species are equally
dispersed in trait space

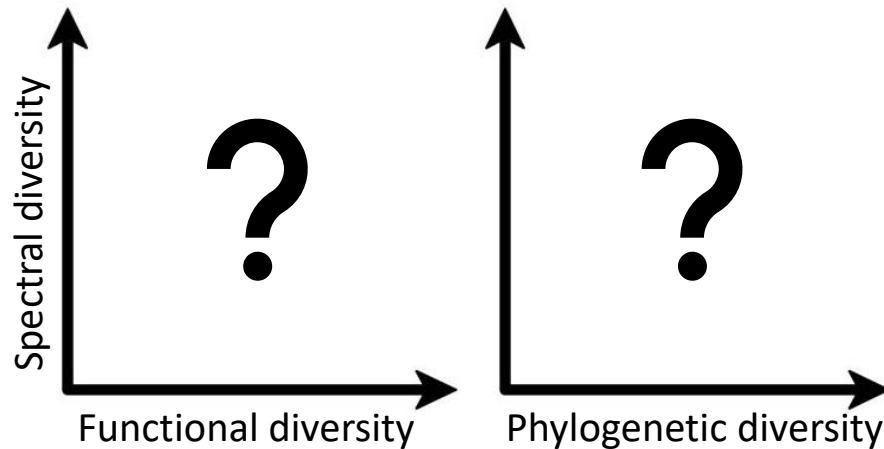
S = species richness
= number of distinct species,
adjusted by their relative
abundance

Distances in
Trait Space

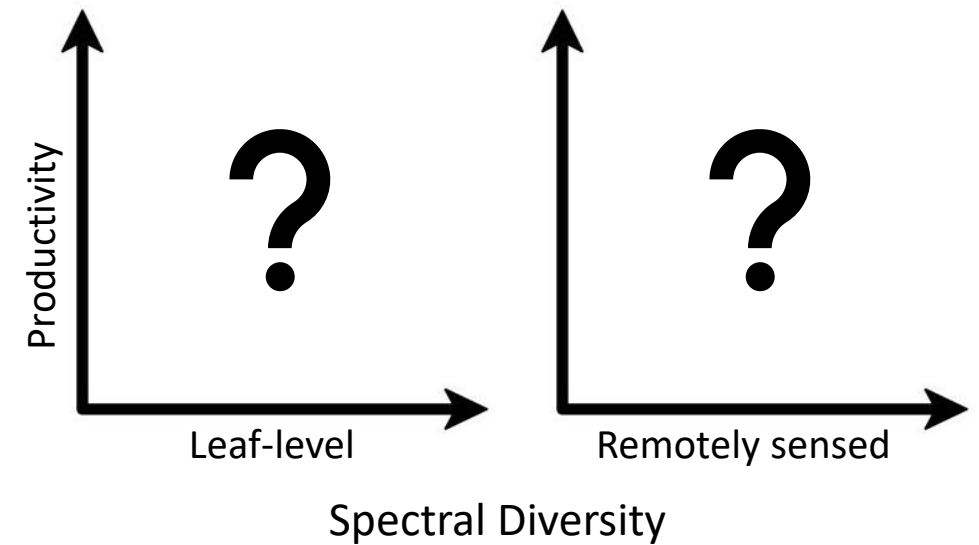
Dissimilarity
Matrix

Methods: Plotting & Statistical Significance

1. Compare Diversity Metrics

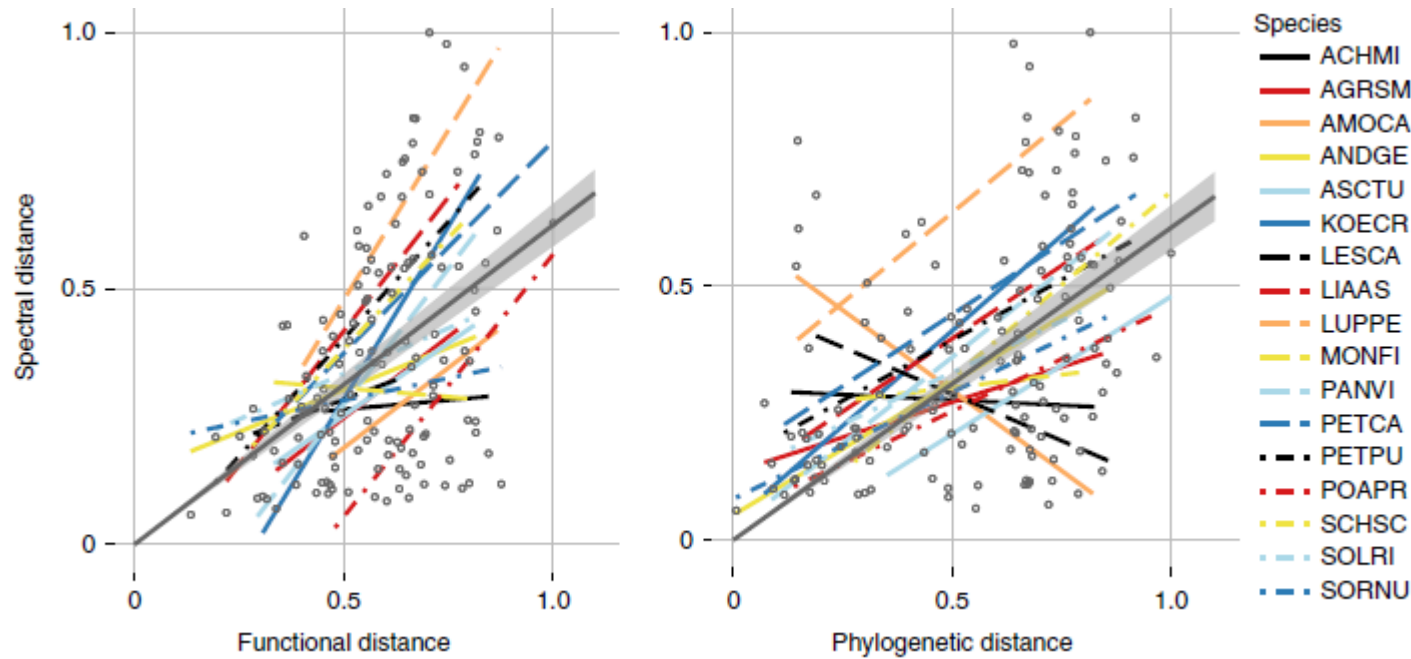


2. Predicting Ecosystem Function



Results

1. Spectral diversity within species of plants **increases** with **functional difference** and **phylogenetic distance**



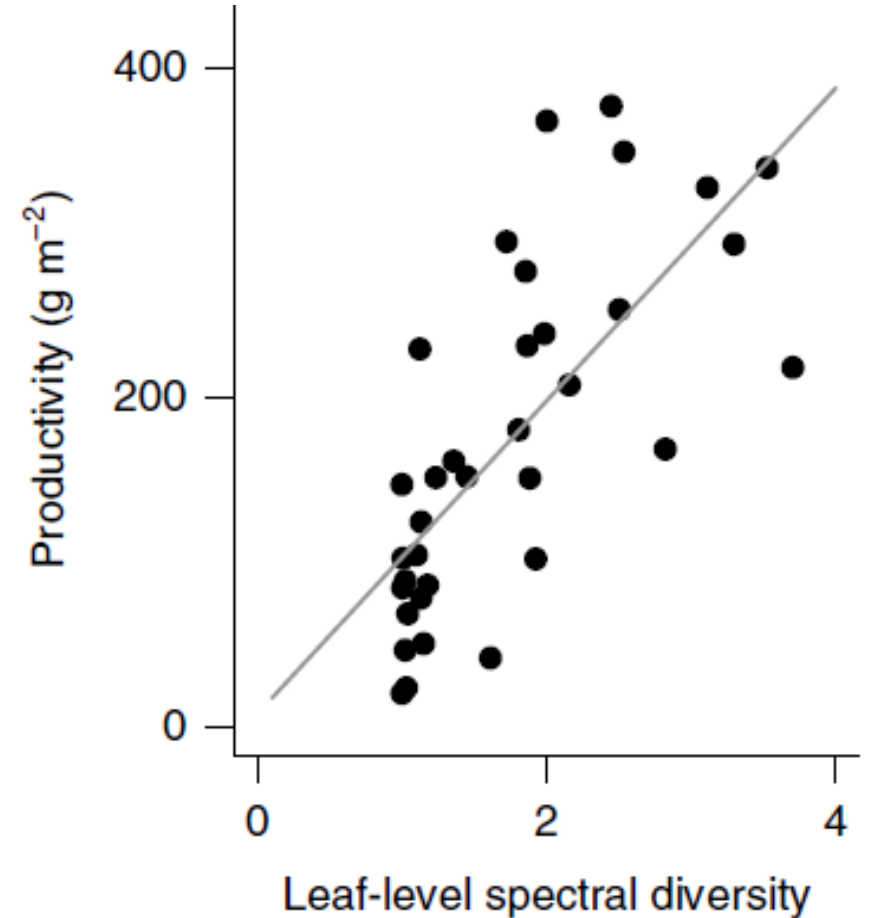
➤ $(r) = 0.42$
➤ $P = 0.001$

(Schweiger et al., 2018, Figure 1)

Results

2. Leaf-level spectral diversity predicted 51% of total **variation** in **productivity**

- $r^2 = 0.51$
- $b = 94.92$
- $t_{33} = 5.90$
- $P < 0.001$;



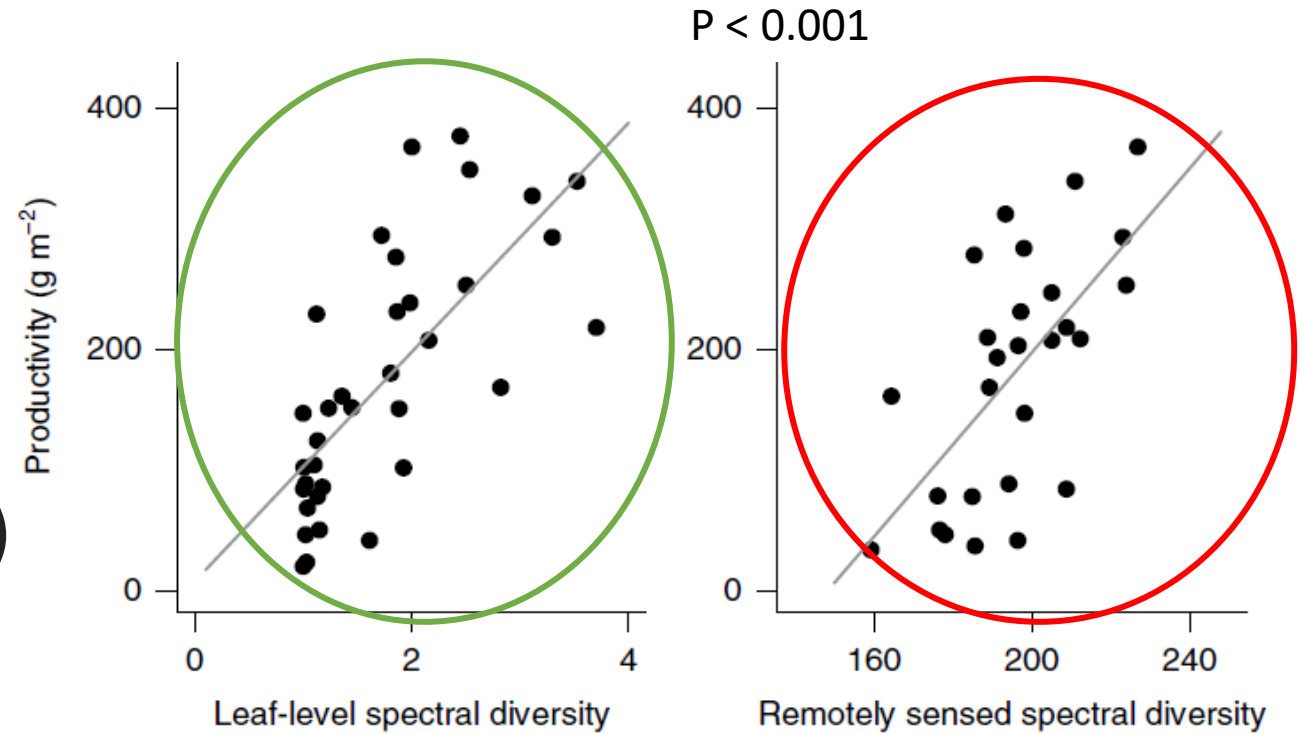
(Schweiger et al., 2018, Figure 3)

Results

2. **Leaf-level** spectral diversity predicted 51% of total variation in productivity

VS.

- Taxonomic (47%)
- Functional (51%)
- Phylogenetic (48%)
- **Remotely sensed** (41%)



(Schweiger et al., 2018, Figure 3)

Key Takeaways



REFLECTING CURRENT SCIENCE

- Spectral diversity reflects biological diversity
- Spectral diversity predicts ecosystem function

Influence of Phylogeny on Fungal Community Assembly and Ecosystem Functioning

July 2007 · [Science](#) 316(5832):1746-8

DOI:[10.1126/science.1143082](#)

Source · [PubMed](#)

Authors:



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(2007)

Evolutionary history and the effects of Biodiversity on plant productivity

December 2008 · [Proceedings of the National Academy of Sciences](#) 105(44):17012-7

DOI:[10.1073/pnas.0805962105](#)

Source · [PubMed](#)

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Bradley J Cardinale



Todd Oakley
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(2008)

Key Takeaways



POTENTIAL ISSUES

- Spectral profiles have limitations
- Remote sensing methods may not be generalizable



NEW APPROACH

- Remote sensing can be acquired rapidly and consistently over large spatial scales

